B28: SQL SuperHero Program : Session 1

1 message

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Hi All,

Please find the below notes from all what we discussed in our today's meeting. Notes will be provided after every class.

For live sql practice, if you do not have tool:

https://livesql.oracle.com/apex/f?p=590:1:28733181935303::NO:::

<Create or account here and use this free tool with latest oracle version available here>

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Tables are basic fundamental storage structure of databases. table which stores data in it. They stored data in rows and columns. Mainly 4 types of tables:

1. Heap Organized
2. Index Organized
3. External Tables
4. Temporary Tables

by default all tables in oracle are heap organized tables. It means oracle inserts a new row, when it gets new data. a new row inserted every time.
Index organized means, everything go on correct place. insert may took costly but selection is easy.

Why Use Index Organized Tables
Accessing data via the primary key is quicker as the key and the data reside in the same structure. There is no need to read an index then read the table data in a separate structure.
Lack of duplication of the key columns in an index and table mean the total storage requirements are reduced.
An index-organized table (IOT) is a type of table that stores data in a B*Tree index structure.
Normal relational tables, called heap-organized tables, store rows in any order (unsorted).

Creation Of Index Organized Tables
To create an index organized table you must:

- Specify the primary key using a column or table constraint.
- Use the ORGANIZATION INDEX.

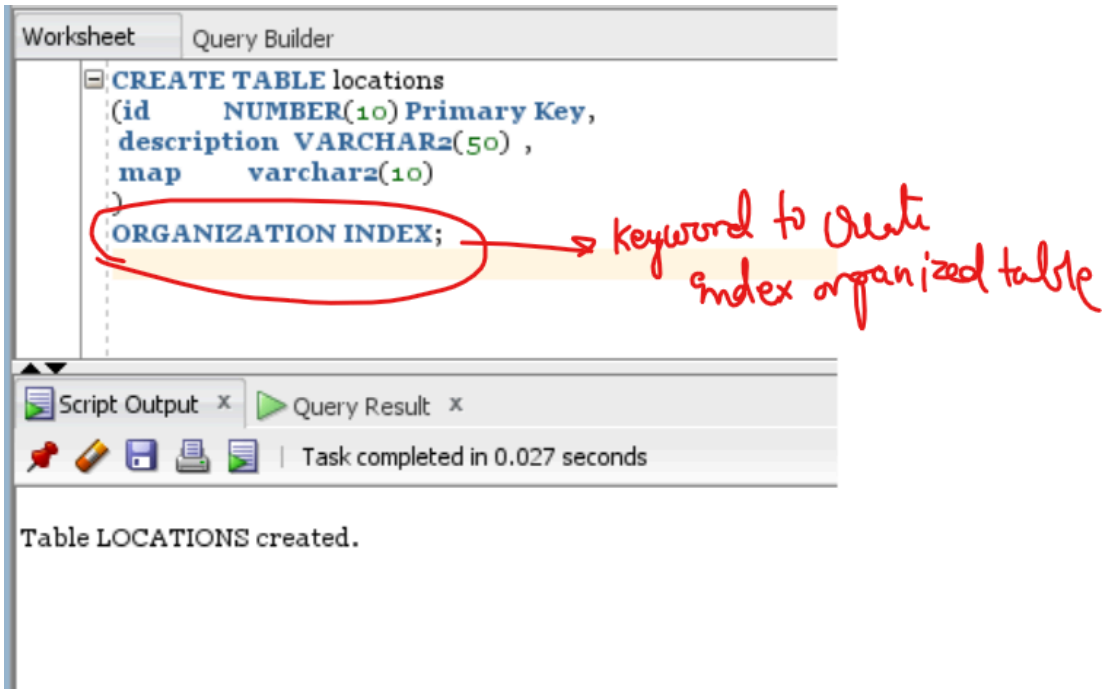
Advantage:

An index-organized table keeps its data sorted according to the primary key column values for the table. An index-organized table stores its data as if the entire table was stored in an index. Indexes serve two main purposes:

- To enforce uniqueness When a PRIMARY KEY or UNIQUE constraint is created, Oracle creates an index to enforce the uniqueness of the indexed columns.
- To improve performance When a query can use an index, query performance may dramatically improve.

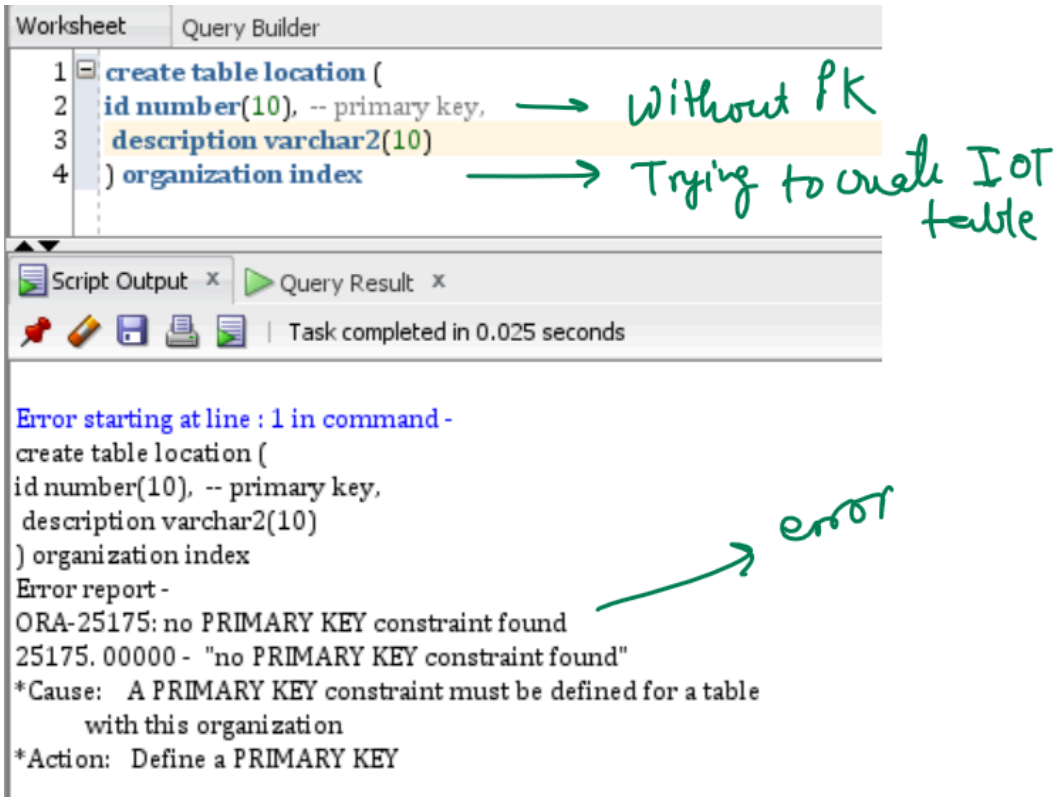
Disadvantage:

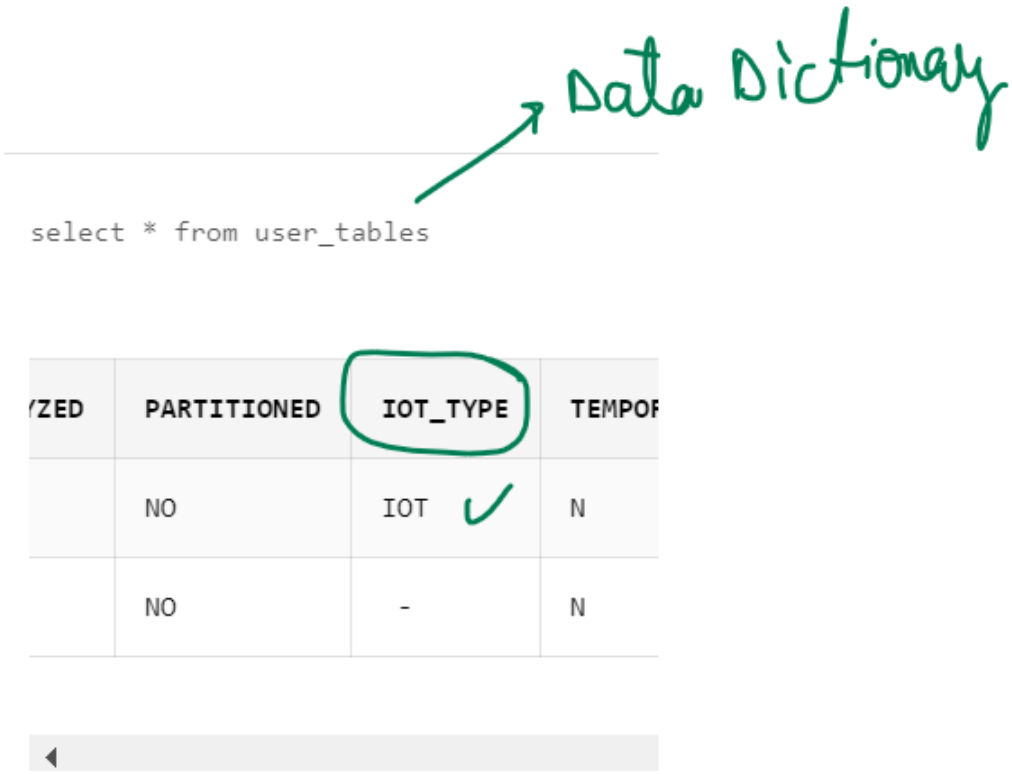
- You must have a primary key on the table with a unique value.
- You cannot have any other indexes on the data.
- You cannot partition an index-organized table.



An index-organized table (IOT) is a type of table that stores data in a B*Tree index structure. Normal relational tables, called heap-organized tables, store rows in any order (unsorted). In contrast to this, index-organized tables store rows in a B-tree index structure that is logically sorted in primary key order.

--> For Creating an IOT(index-organized table), you must need a PK. without that it gives error.





If we check the data dictionary (User_tables) and see the column IOT_TYPE , so we can find the IOT table have the value IOT. Which identifies that this table is IOT table.

--> By default IOT tables are sorted. See the easy example of these both tables as below:

```
-- Insert in Heap organized table (Normal tables)
insert into test values (1,'one');
insert into test values (3,'three');
insert into test values (4,'four');
insert into test values (2,'two');

----------------------------
-- Insert in Index organized table (IOT tables)

insert into location values (1,'one');
insert into location values (3,'three');
insert into location values (4,'four');
insert into location values (2,'two');
```

```
select * from test
```

ID	DESCRIPTION
1	one
3	three
4	four
2	two

```
4 rows selected.
```


```
select * from location
```

ID	DESCRIPTION
1	one
2	two
3	three
4	four

```
4 rows selected.
```

Types of SQL Statement :

- **DDL** --> create, alter, drop, truncate
  - No commit or rollback . It is autocommit.
  - DDL statements are used to build and modify the structure of your tables and other objects in the database. When you execute a DDL statement, it takes effect immediately.
- **DML** --> Insert, update, delete
  - DML statements are used to work with the data in tables.
- **TCL** --> commit, rollback, savepoint

COMMIT: Commit command is used to permanently save any transaction into the database.

ROLLBACK: This command restores the database to last committed state. It is also used with savepoint command to jump to a savepoint in a transaction.

SAVEPOINT: Savepoint command is used to temporarily save a transaction so that you can rollback to that point whenever necessary.

```
select * from mycity;
```

Query Result | SQL | All Rows Fetched: 4 in 0.003 s

	ID	NAME
1	1	UK
2	2	Mumbai
3	3	Chennai
4	4	Kolkatta

```
update mycity set name = 'Noida' where id = 1;
savepoint A;
update mycity set name = 'Noida' where id = 3;
savepoint B;
```

2 save point created

2 rows updated

Query Result | Script Output

Task completed in 0.019 seconds

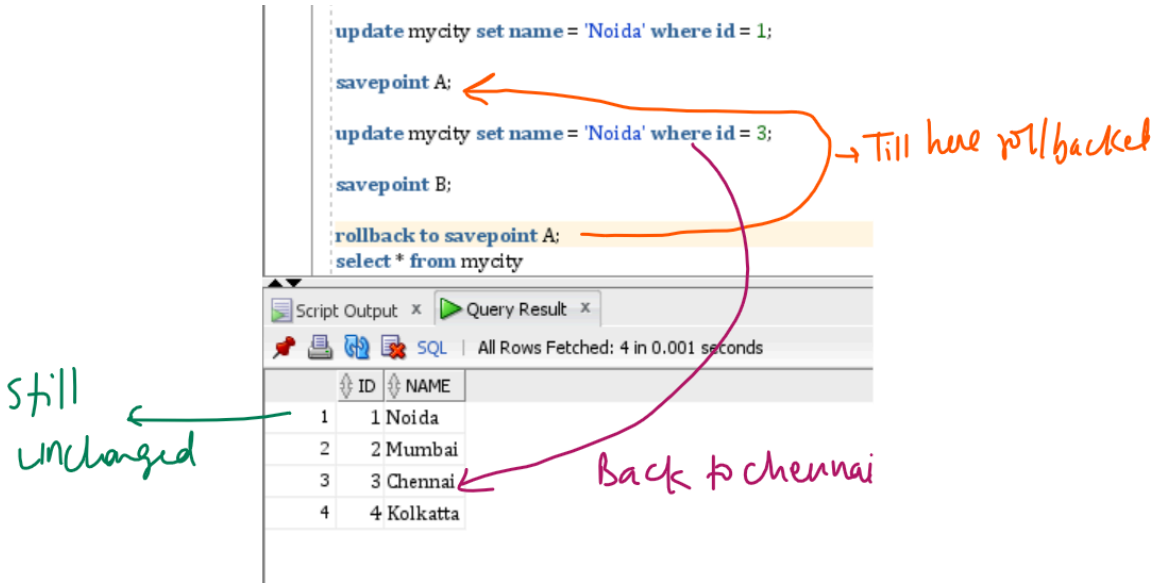
```
1 row updated.

Savepoint created.

1 row updated.

Savepoint created.
```

Now, I can go and revert back until to a particular point and rest was still unchanged.



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Difference between DDL and DML:

DDL	DML
It stands for Data Definition Language.	It stands for Data Manipulation Language.
It is used to create database schema and can be used to define some constraints as well.	It is used to add, retrieve or update the data.
It basically defines the column (Attributes) of the table.	It add or update the row of the table. These rows are called as tuple.
It doesn't have any further classification.	It is further classified into Procedural and Non-Procedural DML.
Basic command present in DDL are CREATE, DROP, RENAME, ALTER etc.	BASIC command present in DML are UPDATE, INSERT, MERGE etc.
DDL does not use WHERE clause in its statement.	While DML uses WHERE clause in its statement.

Question: I am altering a table and adding a column. Now I want to remove it. Can I do rollback?
No, DDL is autocommit, not required commit or rollback

One more question on DDL command:

I have a table Test and that table have id column.

Step 1: Inserted records with id = 1

Step 2: Inserted records with id = 2
Step 3: Create Index on another table Test2
Step 4: Rollback;

How many records in table test as we did the rollback.

Answer: 2 records as we inserted on Step 1 and step 2. because on Step 3 we fired DDL and this is the property of a DDL command to do auto commit of everything what is happened before that DDL.

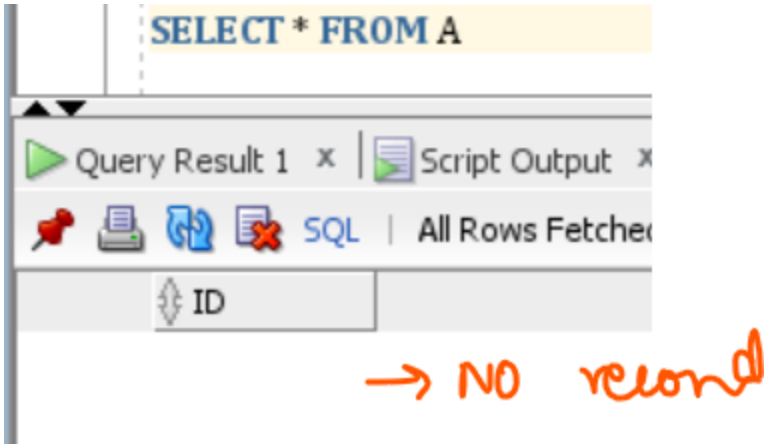
CREATE TABLE A (ID NUMBER) ; -- New table created with 0 records

Scenarion 1:

/* 4 rows Inserted */
INSERT INTO A VALUES (1);
INSERT INTO A VALUES (2);
INSERT INTO A VALUES (3);
INSERT INTO A VALUES (4);

Rollback;

Rollback issues, means everything reverted back.

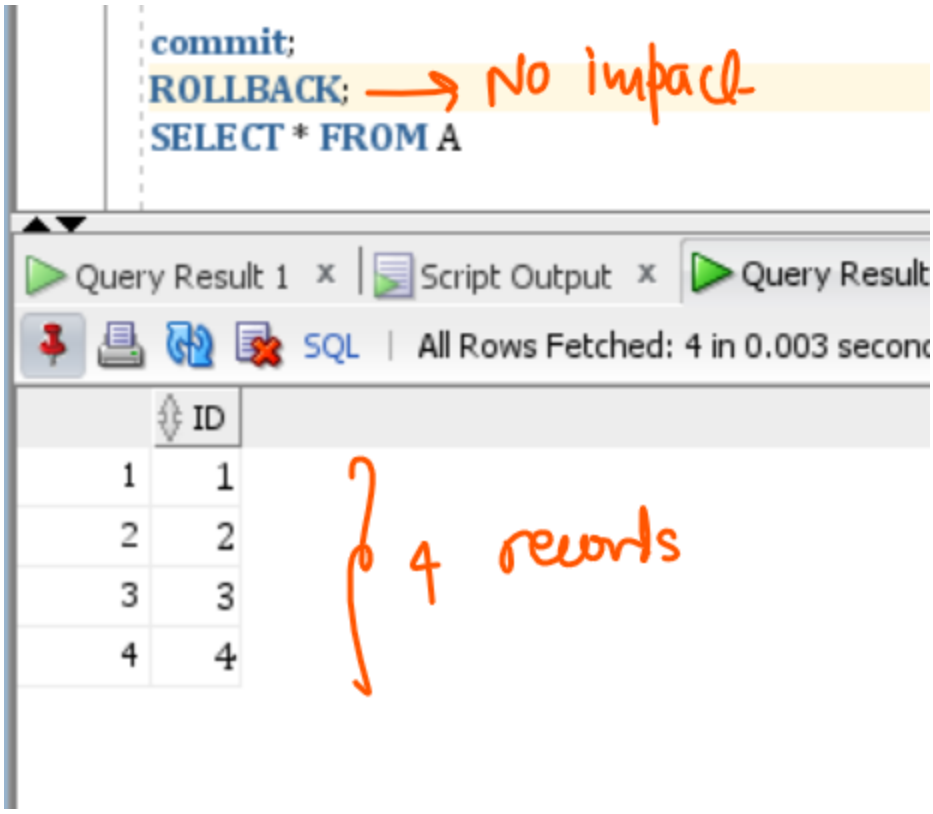


Scenarion 2:

/* 4 rows Inserted */
INSERT INTO A VALUES (1);
INSERT INTO A VALUES (2);
INSERT INTO A VALUES (3);
INSERT INTO A VALUES (4);

Commit;
Rollback;

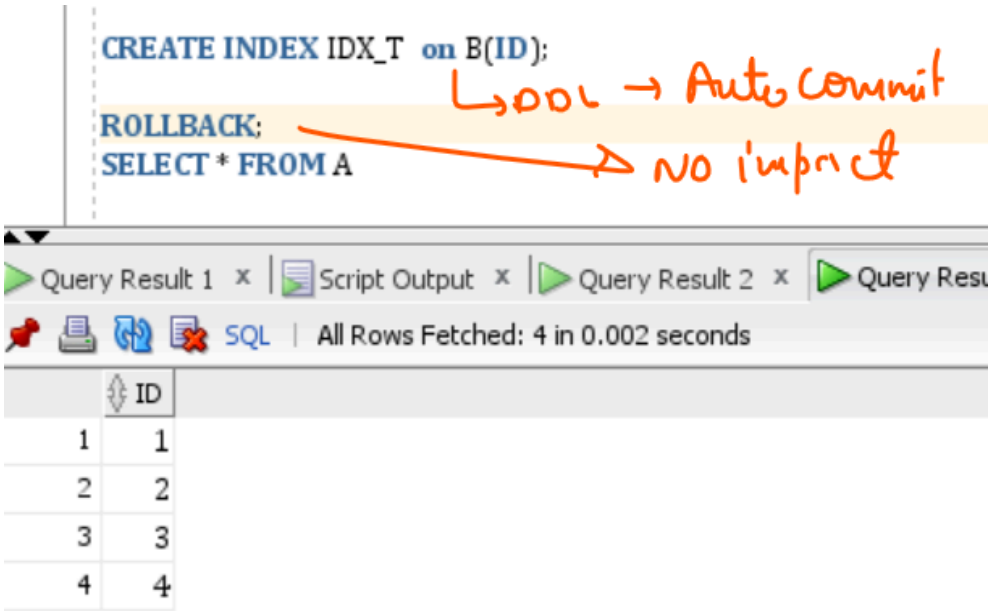
Once something is committed, means committed, you can not rollback. Here there is no impact of rollback.



Scenarion 3:

/* 4 rows Inserted */
INSERT INTO A VALUES (1);
INSERT INTO A VALUES (2);
INSERT INTO A VALUES (3);
INSERT INTO A VALUES (4);

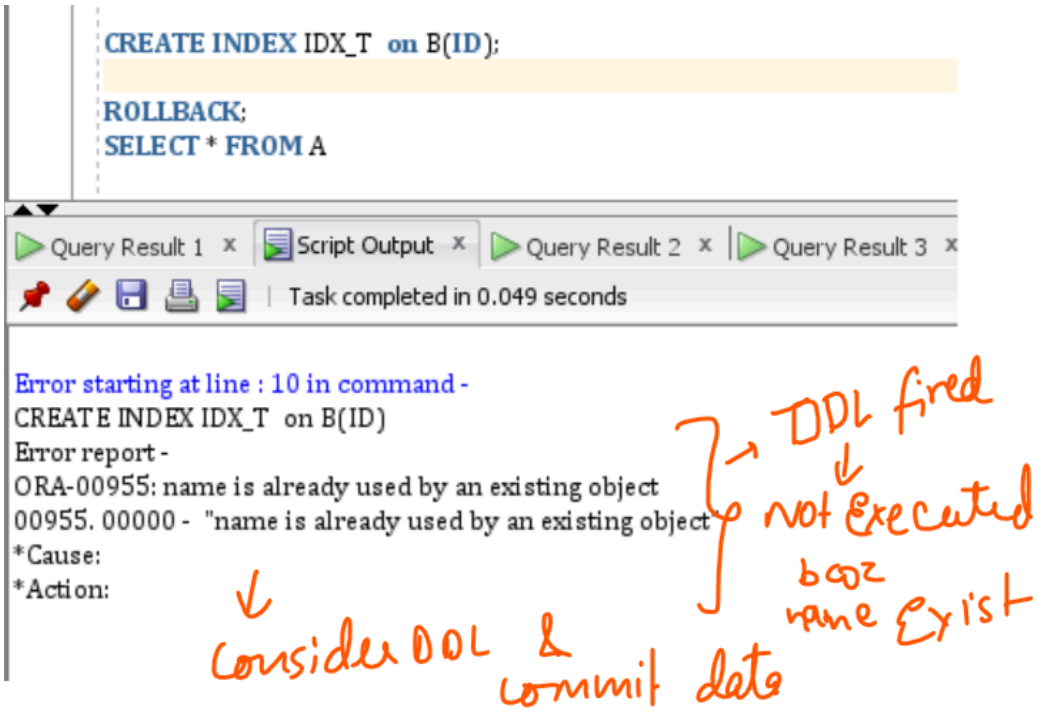
After that I issued a DDL command. DDL can be anything, alter, truncate, drop, create sequence, create index anything



Scenarion 4:

/* 4 rows Inserted */
INSERT INTO A VALUES (1);
INSERT INTO A VALUES (2);
INSERT INTO A VALUES (3);
INSERT INTO A VALUES (4);

After that I issued a DDL command. but my ddl fail.

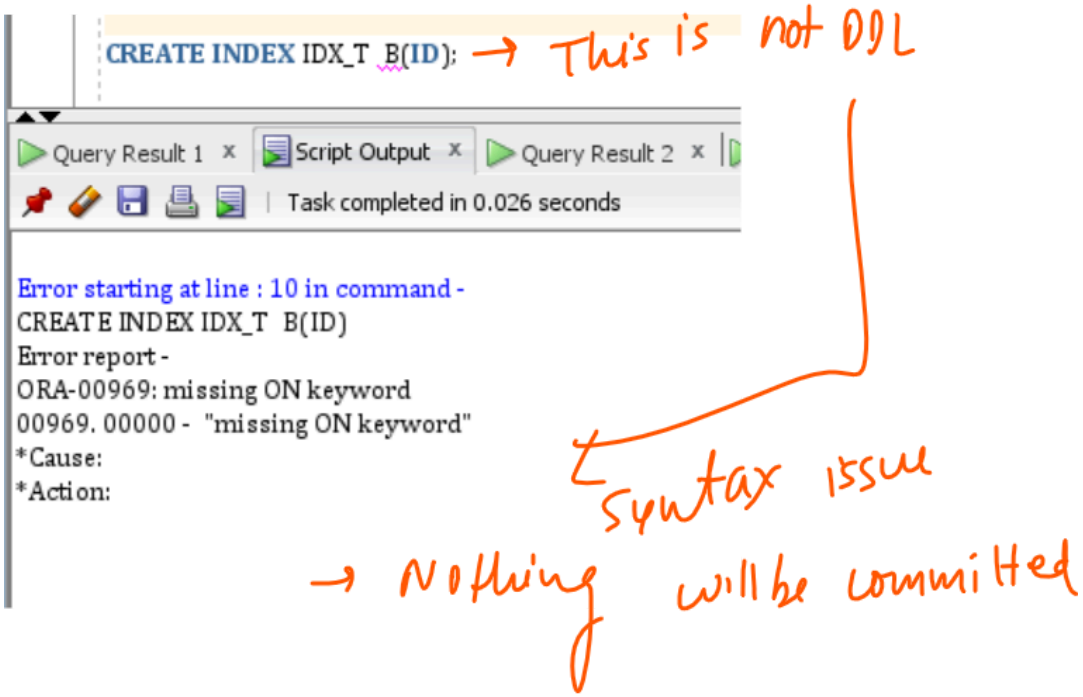


You can see in above example, DDL failed, but still data was inserted in table. Because the concept of DDL is first fire commit internally and then run itself. This question will be twist in interview , when they said, what happened if DDL fails. Remember, if DDL fails (Other than syntax issue) , it commits.

Scenarion 5:

/* 4 rows Inserted */
INSERT INTO A VALUES (1);
INSERT INTO A VALUES (2);
INSERT INTO A VALUES (3);
INSERT INTO A VALUES (4);

After that I issued a DDL command. but my ddl fail, but it should be syntax error. SYNTAX error means, database not consider that you are a DDL. data will not commit.



But If DDL fails because of Syntax issue, means database does not consider it as DDL, it will not commit.

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Datatypes:

```
create table employee(
EMPID number,
EMPNAME varchar2(10),
age number,
sex char(4)
);
```

varchar is for string.
number is for number.

varchar2(10) --> 10 is the capacity of that column. it can maimum store the 10 character string.

--> syntax --> means how the programming language talk to system.

--> **Insert syntax**
--> Insert Into <tablename> Values(the value you want to insert);

insert into employee values (1,'hsgsggs',100,'M');

select * from employee; * means we want to see all data on the table.

If I am trying to insert values more than 10 character like:

insert into employee values (1,'hsgsyeyyeetggs',100,'M');

It gives me an error, because the capacity of the empname column is 10 and we are trying to insert more than 10 character strings.

varchar2(10) --> indicates the total capacity of this column is 10. we can not insert any data which length is more than 10

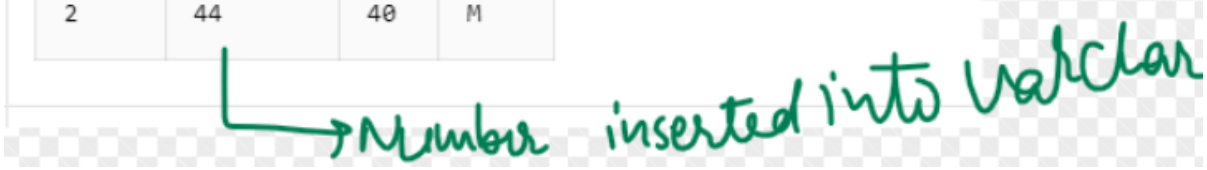
--> In varchar2 columns we can insert number values , but in number columns we can not insert varchar2 values

```
insert into emp values (2, 44, 40, 'M')
```

```
1 row(s) inserted.
```

```
SELECT * FROM EMP WHERE emapname = '44'
```

EMPID	EMAPNAME	AGE	SEX
2	44	40	M




```
SELECT * FROM EMP WHERE emapname = '44'
```

EMPID	EMAPNAME	AGE	SEX
2	44	40	M

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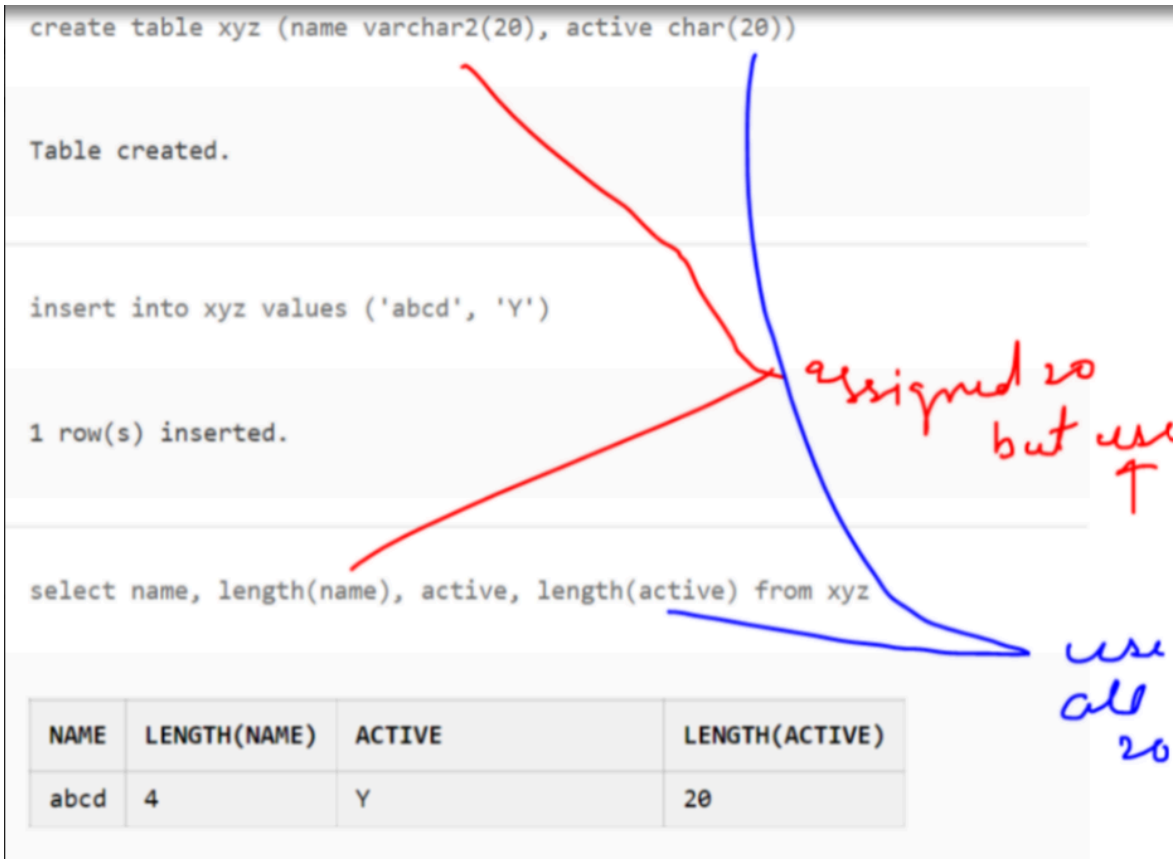

Difference between char and varchar ?

The short answer is: VARCHAR is variable length, while CHAR is fixed length. CHAR is a fixed length string data type, so any remaining space in the field is padded with blanks. CHAR takes up 1 byte per character. ... VARCHAR is a variable length string data type, so it holds only the characters you assign to it.

Prefer to use varchar2

Difference between char, varchar and VARCHAR2 in Oracle :

Sno	Char	VarChar/VarChar2
1	Char stands for **"Character"**	VarChar/VarChar2 stands for **Variable Character**
2	It is used to store character string of **fixed length**	It is used to store character string of **variable length**
3	It has a Maximum Size of **2000** Bytes	It has a Maximum Size of **4000** Bytes

Sno	Char	VarChar/VarChar2
4	Char will pad the spaces to the right side to fill the length specified during the Declaration	VarChar will not pad the spaces to the right side to fill the length specified during Declaration.
6	It is Static Datatype(i.e Fixed Length)	It is Dynamic Datatype(i.e Variable Length)
7	It can lead to memory wastage	It manages Memory efficiently
8	It is 50% much faster than VarChar/VarChar2	It is relatively slower as compared to Char

Note : There is no difference between VarChar and VarChar2 in Oracle. However, it is advised not to use VarChar for storing data as it is reserved for future use for storing some other type of variable. Hence, always use VarChar2 in place of VarChar.

Hence it is advised to use Char datatype when the length of the character string is fixed and will not change in the future.
If the length of the character string is not fixed, then VarChar2 is preferred